

Models for Exceedances over High Thresholds

By A. C. DAVISON†

and

R. L. SMITH

University of Oxford, UK

University of Surrey, Guildford, UK

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SUMMARY

We discuss the analysis of the extremes of data by modelling the sizes and occurrence of exceedances over high thresholds. The natural distribution for such exceedances, the generalized Pareto distribution, is described and its properties elucidated. Estimation and model-checking procedures for univariate and regression data are developed, and the influence of and information contained in the most extreme observations in a sample are studied. Models for seasonality and serial dependence in the point process of exceedances are described. Sets of data on river flows and wave heights are discussed, and an application to the siting of nuclear installations is described.

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1. INTRODUCTION

Extreme value theory has developed very rapidly in recent years. Galambos's (1978) monograph was followed by more specialized works by Leadbetter *et al.* (1983), Resnick (1987) and Reiss (1989), the NATO Advanced Study Institute volume edited by Tiago de Oliveira (1984) and a host of related publications. Nevertheless, many statisticians probably share Loynes's (1986) misgivings about the lack of practical statistical methodology. In this paper we propose one version of such a methodology, in which statistical models are fitted to exceedances over high thresholds.

The problem may be described, in its broadest terms, as how to make statistical inferences about the extreme values in a population or a random process. The traditional approach (Gumbel, 1958) is based on the extreme value limit distributions first identified by Fisher and Tippett (1928). There is an enormous literature on this approach, applied both to environmental data such as river heights, sea levels, rainfall and air pollutants, and also in the reliability context where the corresponding distributions for sample minima (in particular, the Weibull distribution) are used to model strengths and failure times. A typical application consists of the generalized extreme value distribution fitted to annual maxima. This is defined by

$$H(y; \mu, \psi, k) = \exp \left\{ - \left(1 - k \frac{y - \mu}{\psi} \right)^{1/k} \right\} \quad (1.1)$$

†*Address for correspondence:* Department of Statistics, University of Oxford, 1 South Parks Road, Oxford, OX1 3TG, UK.

over the range of y that $k(y - \mu) < \psi$, where $\psi > 0$ and k, μ are arbitrary. The case $k = 0$ is interpreted as the limit $k \rightarrow 0$, i.e.

$$H(y; \mu, \psi, 0) = \exp \left\{ - \exp \left(- \frac{y - \mu}{\psi} \right) \right\}, \quad (1.2)$$

the type I family of Gumbel (1958), sometimes called the Gumbel distribution. These distributions are justified as the limiting stable distributions of extreme value theory. The *Flood Studies Report* (National Environment Research Council, 1975) recommended this approach to extreme value problems in hydrology and proposed numerical fitting procedures. The numerical aspects were developed by Prescott and Walden (1980, 1983) and a Newton–Raphson algorithm published by Hosking (1985) and amended by Macleod (1989). For the asymptotics of maximum likelihood, see Smith (1985); the most important thing to note is that the regularity conditions are satisfied when $k < 0.5$ and in that case the usual properties of consistency, asymptotic efficiency and asymptotic normality hold. Nevertheless there is evidence that maximum likelihood estimation does not work well in small samples, and other methods have been proposed, the most serious competitor being the ‘probability weighted moments’ approach advocated by Hosking *et al.* (1985).

An extension of the classical approach is to use, instead of just annual maxima, several of the largest order statistics in each year. An appropriate class of limiting distributions is known as a direct extension of equations (1.1) and (1.2); see for example section 2.3 of Leadbetter *et al.* (1983). This was first proposed as a statistical procedure by Weissman (1978) and developed further by Gomes (1981), Smith (1986) and Tawn (1988). We shall not discuss this idea any further in the present paper.

Threshold methods have been developed by hydrologists over the last 20 years, often under the acronym of the POT (peaks over threshold) method. These methods are based on fitting a stochastic model to either the exceedances or the peaks over a threshold. Such methods in some form have been around for a long time, but the first systematic development was in the works of Todorovic and Zelenhasic (1970) and Todorovic and Rousselle (1971). Early versions assumed a non-homogeneous Poisson process to model the times of exceedances over high thresholds, combined with independent exponential random variables to represent the amounts by which the threshold is exceeded (the excesses). North (1980) proposed a seasonal version in which both the intensity of the process of exceedances and the mean excess level depend on sinusoidal components in time. Numerous more recent variants have been proposed.

Our own approach is based on the generalized Pareto distribution, which has an interpretation as a limiting distribution similar to that which motivates equation (1.1). This idea was first suggested by Pickands (1975) and has been developed by, amongst others, DuMouchel (1983), Davison (1984a, b), Smith (1984, 1987), van Montfort and Witter (1985, 1986), Hosking and Wallis (1987) and Joe (1987).

Section 2 defines the generalized Pareto distribution and describes some of its basic statistical properties. Section 3 is concerned with regression models in which one or both of the generalized Pareto parameters depend on covariates. Both theoretical and computational aspects are discussed. Section 4 is concerned with diagnostics, and Section 5 with the question of the influence of extreme observations—a vital point in any discussion of ‘robust’ methods in this context. Section 6 provides an application of regression modelling, concerned with the spread of contaminants from nuclear

installations. The remaining sections are concerned with applications to environmental time series in which both serial dependence and seasonality must be taken into account. Section 7 provides theoretical background on these two topics and Sections 8 and 9 discuss two examples in the hydrology/oceanography sphere. Our conclusions are summarized in Section 10. Parts of the paper review and extend ideas presented in cruder form by Davison (1984b) and Smith (1984).

2. THE GENERALIZED PARETO DISTRIBUTION

The simplest models for exceedances over thresholds are based on a Poisson process of exceedance times combined with independent excesses over the threshold. We consider this case first before generalizing to more complicated models in later sections.

Fix a threshold u , and let N denote the number of exceedances of u during a period of m years. In accordance with most environmental applications, we take 1 year as the unit of time, though this is of course not necessary for the theory. Suppose the excesses, or differences between the observations over the threshold and the threshold itself, are independent with common distribution function G . The theory of this paper is built around the assumption that G is the generalized Pareto distribution given by

$$G(y; \sigma, k) = 1 - \left(1 - k \frac{y}{\sigma}\right)^{1/k} \quad (2.1)$$

where $\sigma > 0$ and k is arbitrary; the range of y is $0 < y < \infty$ if $k \leq 0$, $0 < y < \sigma/k$ if $k > 0$. The case $k = 0$ is interpreted as the limit $k \rightarrow 0$, i.e. the exponential distribution with mean σ . The case $k < 0$ is just a reparameterization of one of several forms of the usual Pareto distributions, but the extension to $k \geq 0$ was given by Pickands (1975).

There are several ways of motivating equation (2.1). Pickands showed that equation (2.1) arises as a limiting distribution for excesses over thresholds if and only if the parent distribution is in the domain of attraction of one of the extreme value distributions. A more direct motivation is the 'threshold stability' property: if Y is generalized Pareto and $u > 0$, then the conditional distribution of $Y - u$ given $Y > u$ is also generalized Pareto. Another property is the following: if N has a Poisson distribution and, conditionally on N , $Y_1, \dots, Y_N$ are independent identically distributed generalized Pareto random variables, then $\max(Y_1, \dots, Y_N)$ has the generalized extreme value distribution (see equation (9.1) following). Thus, a Poisson process of exceedance times with generalized Pareto excesses implies the classical extreme value distributions. Both the last two properties characterize the generalized Pareto distribution in the sense that no other family has either property.

These properties suggest that the generalized Pareto distribution will be a practical family for statistical estimation, provided that the threshold is taken sufficiently high. An important question is how high to take the threshold. Theoretical results are available (Smith, 1987) but these have not been taken sufficiently far to be of practical value. Instead, we shall adopt a more pragmatic approach in which a range of diagnostic techniques is used to assess the fit of the model.

Methods for estimating the generalized Pareto distribution parameters have been reviewed by Hosking and Wallis (1987). Maximum likelihood estimators exist in large samples provided that $k < 1$ and are asymptotically normal and efficient if $k < \frac{1}{2}$ (Smith, 1985). The limiting covariance matrix is obtained from the inverse of the Fisher information matrix as

$$\frac{1-k}{N} \begin{pmatrix} 2\sigma^2 & \sigma \\ \sigma & 1-k \end{pmatrix} \quad (2.2)$$

provided that $k < \frac{1}{2}$. It is possible to reduce maximum likelihood estimation to a one-dimensional search, as follows. The log-likelihood based on $Y_1, \dots, Y_N$ is

$$-n \log \sigma + (1/k - 1) \sum_{i=1}^N \log(1 - k Y_i / \sigma).$$

Writing $\sigma = k/\tau$ and differentiating with respect to τ and k , we see that the maximum with respect to k is achieved when

$$k = k(\tau) = -N^{-1} \sum_{i=1}^N \log(1 - \tau Y_i).$$

Thus maximum likelihood estimation reduces to solution of the single equation

$$N/\tau = \{1/k(\tau) - 1\} \sum_{i=1}^N Y_i / (1 - \tau Y_i).$$

Further numerical aspects are considered in Section 3 and Appendix B.

Among other methods of estimation, the main competitor to date is the probability weighted moments method of Hosking and Wallis (1987). The r th probability weighted moment is defined to be

$$\alpha_r = E[Y\{1 - G(Y)\}^r] = \frac{\sigma}{(r+1)(r+1+k)},$$

where G is the generalized Pareto distribution function. The method essentially consists of equating the theoretical and sample-based values of the probability weighted moments for $r=0, 1$. Hosking and Wallis gave formulae for the approximate standard errors of the estimates so obtained.

A graphical method of estimation is suggested by the following relation. If $k > -1$, $u > 0$ and $\sigma - uk > 0$, then

$$E(Y - u | Y > u) = \frac{\sigma - ku}{1 + k},$$

which suggests a 'mean residual life' plot (Yang, 1978; Hall and Wellner, 1981) in which the mean observed excess over u is plotted against u . If the generalized Pareto assumption is correct then the plot should follow a straight line with intercept $\sigma/(1+k)$ and slope $-k/(1+k)$; this suggests both graphical estimates of σ and k and a test of fit based on the linearity of the plot.

It is also necessary to estimate the crossing rate of the threshold. If the exceedance process is assumed to be Poisson with rate λ (per year), then an unbiased estimate of λ is N/m , with estimated variance N/m^2 .

In environmental studies the most common quantity of interest is the n -year return level, defined (loosely) as that level which is exceeded on average once in n years.

A more precise definition for our model is as follows: since for $x > u$ the mean crossing rate (per year) of level x is $\lambda\{1 - k(x - u)/\sigma\}^{1/k}$, we set this equal to $1/n$ to obtain

$$q_n = u + \sigma \frac{1 - (\lambda n)^{-k}}{k}. \quad (2.3)$$

A point estimate may be obtained by substituting the maximum likelihood or other estimates into equation (2.3). Interval estimates may be obtained by the delta method (Rao (1973), p. 385), using either equation (2.2) or the observed information matrix with maximum likelihood, or the analogue of equation (2.2) with probability weighted moments estimates (equation (6) of Hosking and Wallis (1987)). An alternative procedure is to reparameterize the problem in terms of (q_n, k, λ) , treating k and λ as nuisance parameters and using equation (2.3) to solve for σ , and then to construct a profile likelihood for q_n . The potential advantage of this, as will be seen in Section 9, is that the confidence intervals so constructed tend to be highly asymmetric, a property which the delta method often does not capture adequately even after a parameter transformation.

3. REGRESSION

A major theme of statistics is the description of systematic variation in a variable of primary interest, a response, in terms of covariates. We may wish to combine information from different but related sets of data or to adjust for seasonal variation. Furthermore the generalized Pareto approximation to the distribution of excesses improves at the higher thresholds. By combining several data sets using covariates, it is possible to use a higher threshold and thus to improve the fit of the model.

We suppose below that k and σ are functions of covariates and regression parameters. In view of the requirement $\sigma > 0$, one natural formulation is $\sigma(\mathbf{x}^T \boldsymbol{\beta}) = \exp(\mathbf{x}^T \boldsymbol{\beta})$, but our discussion is more general. In generalized linear models it is natural to model variation in terms of relationships between a linear predictor, the mean, and possibly a dispersion parameter. In the statistics of extremes, however, distribution moments do not exist for all values of k , and the direct interpretations of k and σ in terms of distribution tail behaviour suggest that a more fruitful approach is to model variation in σ and occasionally k . So far we have found this to be satisfactory.

3.1. Maximum Likelihood

Suppose that the Y_i , $i = 1, \dots, N$ are independent generalized Pareto random variables with parameters $k_i = k(\mathbf{z}_i^T \boldsymbol{\gamma})$ and $\sigma_i = \sigma(\mathbf{x}_i^T \boldsymbol{\beta})$, where $\mathbf{x}_i$ and $\mathbf{z}_i$ are known vectors of covariates of respective lengths $p + 1$ and $q + 1$. The parameters β_0 and γ_0 will represent overall 'means', so we take $x_{i0} = z_{i0} = 1$. By analogy with a simple random sample, standard asymptotics of maximum likelihood estimation apply if each $k_i < \frac{1}{2}$, as we henceforth assume. The methodology extends directly to the case where the data are right censored. Let $\delta_i = 0$ if the value of Y_i is observed, and $\delta_i = 1$ if the observed value y_i is a lower bound for the actual Y_i . This is rather a general set-up for modelling: in most applications there will be no censoring and each k_i will

equal a common k . Formulae for the log-likelihood L , the score statistic and observed information I_θ and expected information i_θ are given in Appendix A.

When there is a single value of k , fitting can be performed by iterative weighted least squares. The methods of Green (1984) or Jørgensen (1984) can be used to estimate β at each k over a suitable range. Expected information weights should be used in the fitting algorithm because if $-1 \leq k \leq 0.5$ there is appreciable probability (0.3–0.5) that the observed information for an uncensored observation, $-\partial^2 l_i / \partial \sigma_i^2$, is negative. Appendix B outlines how the calculation may be performed in GLIM. Presumably different values of k could be accommodated by use of a seesaw algorithm such as that of Aitkin (1987), but we have not pursued this. Alternatively, standard Newton–Raphson or quasi-Newton methods may be applied.

The profile log-likelihood for k is usually not quadratic in small and moderate samples, and it provides a better basis for confidence intervals than observed or expected information. Standard errors for β obtained directly from iterative weighted least squares must be corrected to allow for estimation of k . The adjustment used by Aitkin and Clayton (1980) is easily adapted to our setting. If k is constant, there is no censoring, and $\sigma = \exp \eta$, then this difficulty can be avoided by reparameterization so that regression effects are orthogonal to the mean, i.e. $\sum x_{ij} = 0$, $j = 1, \dots, p$. The inverse Fisher information for (k, β) is then

$$\begin{pmatrix} 2(1-k)/N & (1-k)/N & \mathbf{0} \\ (1-k)/N & (1-k)^2/N & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & (1-2k)(\mathbf{X}_{(0)}^T \mathbf{X}_{(0)})^{-1} \end{pmatrix},$$

of which the lower right-hand corner is provided directly by GLIM. Here $\mathbf{X}_{(0)}$ denotes the design matrix without the column corresponding to β_0 .

3.2. Least Squares

When the k_i are all equal and there is no censoring, naïve least squares can be used to estimate k and β . If we suppose again that $\sigma_i = \exp(\beta_0 + \mathbf{x}_{i(0)}^T \beta_{(0)})$, where $\beta_{(0)} = (\beta_1, \dots, \beta_p)^T$, the logarithms $v_i = \ln y_i$ of the data satisfy the linear model

$$\mathbf{v} = \mathbf{X}_{(0)} \beta_{(0)} + (\beta_0 + \kappa_1) \mathbf{j} + \epsilon.$$

Here $\mathbf{j}$ is an $N \times 1$ vector of ones, ϵ an $N \times 1$ vector of independent log-generalized-Pareto random variables centred to have mean zero, $\mathbf{x}_{i(0)}^T$ is the i th row of $\mathbf{X}_{(0)}$, and κ_1 is the mean of the log-generalized-Pareto distribution with $\sigma = 1$. This and other cumulants κ_r are derived from the cumulant generating function

$$K(t) = \begin{cases} t \ln(\sigma/k) + \ln \Gamma(t+1) + \ln \Gamma(1+1/k) - \ln \Gamma(1+t+1/k) & k > 0 \\ t \ln \sigma + \ln \Gamma(t+1) & k = 0 \\ t \ln(-\sigma/k) + \ln \Gamma(t+1) + \ln \Gamma(-t-1/k) - \ln \Gamma(1-1/k) & k < 0, \end{cases}$$

which is defined for $-1 < t < -1/k$ if $k < 0$, and $t > -1$ otherwise. Without loss of generality we parameterize so that the columns of $\mathbf{X}_{(0)}$ sum to zero, i.e. the mean is orthogonal to the regression coefficients.

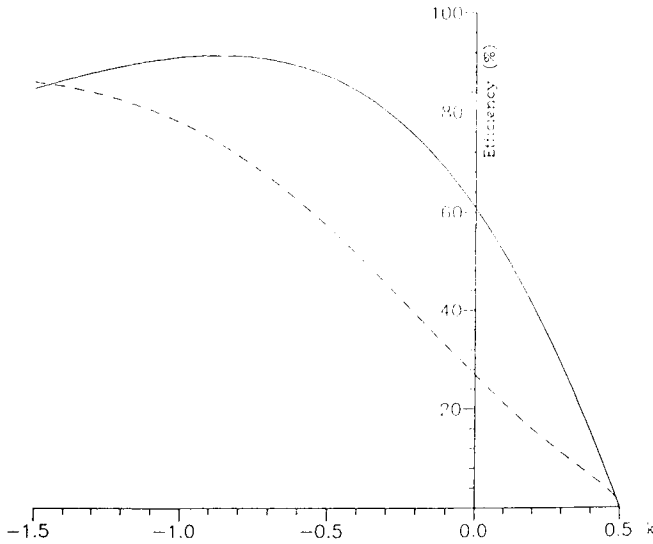


Fig. 1. Efficiency (%) of least squares estimates as functions of k : —, $E_1(k)$; ----, $E_2(k)$

The least squares estimate $\beta_{(0)}^* = (\mathbf{X}_{(0)}^T \mathbf{X}_{(0)})^{-1} \mathbf{X}_{(0)}^T \mathbf{v}$ is an unbiased estimate of $\beta_{(0)}$, with covariance matrix $(\mathbf{X}_{(0)}^T \mathbf{X}_{(0)})^{-1} \kappa_2$ (Cox and Hinkley, 1968). It follows that if $k < \frac{1}{2}$ the asymptotic relative efficiency of $\beta_{(0)}^*$ is $E_1(k) = (1 - 2k)/\kappa_2 \times 100\%$. Estimates of β_0 and k can be obtained from the mean $\bar{v}$ of $v_1, \dots, v_N$ and the residual sum of squares S^2 by solution of the equations

$$S^2 = (N - p) \kappa_2(k)$$

$$\bar{v} = \beta_0 + \kappa_1(k).$$

The joint asymptotic efficiency

$$E_2(k) = \left[\frac{\{(1 - k) d\kappa_2/dk\}^2 (1 - 2k)}{\kappa_4 \kappa_2 + 2\kappa_2^3 - \kappa_3^2} \right]^{1/2} \times 100\%$$

of these estimates is plotted in Fig. 1, together with $E_1(k)$. The calculation to obtain E_2 mimics that of Newby (1980). The overall efficiency of estimation of all the parameters in a model where β has dimension p is $\{E_1(k)^p E_2(k)^2\}^{1/(p+2)}$, which approaches E_1 for large p . Fig. 1 indicates that the loss of information if k and β_0 are estimated is severe if $|k| < \frac{1}{2}$, but that least squares estimation of β is reasonably efficient if $k < 0$. This suggests the following strategy for finding initial values of parameter estimates for use in optimization routines: use least squares to estimate β , then fix β and apply the one-dimensional root search mentioned in Section 2 to the variables $Y_i \exp(-\mathbf{x}_i^T \beta^*)$, thereby estimating k and $\sigma = \exp \beta_0$. The resulting values of $\hat{k}$, $\hat{\beta}_0$ and β^* may be used as initial values in optimization routines.

4. DIAGNOSTICS

Residuals are very useful in assessing the fit of a model to data. There are many definitions of residuals in the literature, among them deviance residuals (Pregibon, 1981) and those due to Cox and Snell (1968). Deviance residuals for the generalized Pareto distribution are defined as

$$\Delta_i = \text{sign}(\tilde{\sigma}_i - \hat{\sigma}_i) [2\{l(y_i; \hat{k}, \tilde{\sigma}_i) - l(y_i; \hat{k}, \hat{\sigma}_i)\}]^{1/2},$$

where $\tilde{\sigma}_i$ is the value of σ which maximizes the log-likelihood contribution $l(y_i; \hat{k}, \sigma)$ due to y_i and $\hat{\sigma}_i$ is the value of σ fitted under the model. In many cases recentred and scaled deviance residuals are very close to standard normal, but Davison (1984a) showed this not to be so for the generalized Pareto distribution. In the statistics of extremes the largest observations have a big influence on the fitted model, so it seems natural to choose a scale for residuals in light of this. This suggests that Cox-Snell residuals on the exponential scale be used. Hence we let $r_i = -\ln(1 - \hat{k}y_i/\hat{\sigma}_i)/\hat{k}$ if $\hat{k} \neq 0$, and $r_i = y_i/\hat{\sigma}_i$ if the model holds $k = 0$. Another reason for this choice is that, if a regression model is fitted to both k and σ , Cox-Snell residuals are well behaved but deviance residuals take values $\pm \infty$. The r_i can be transformed to normality or uniformity if either is thought better for a particular purpose, such as checking for serial correlation. Cox and Snell (1968) showed how to correct the r_i for the effects of estimating the parameters, but their numerical examples suggest that the corrections make only slight differences to residual plots: henceforth we ignore them. Davison and Gigli (1989) compared deviance and Cox-Snell residuals in detail.

We shall see in Section 5 that estimates of the parameters of the generalized Pareto distribution are sensitive to the most extreme observations in a sample. Moreover sample sizes in applications can be small, making measures of the influence of individual observations on parameter estimates essential. Diagnostics can be obtained by applying the case deletion scheme suggested by Cook and Weisberg (1982). Let $\hat{\theta}_{(i)}$ denote the maximum likelihood estimate of θ when the i th observation is deleted. Then a measure of the distance between $\hat{\theta}$ and $\hat{\theta}_{(i)}$ in terms of the overall log-likelihood L is

$$2\{L(\hat{\theta}) - L(\hat{\theta}_{(i)})\}. \quad (4.1)$$

Observations for which this is large, compared with the χ^2_{p+q+2} distribution from which confidence regions for θ are found, are regarded as influential. If we standardize by dividing expression (4.1) by $p+q+2$ we obtain an analogue of the exact Cook statistics used for linear normal models (Cook, 1977). The computational expense of calculating the $\hat{\theta}$ for each observation in turn can be reduced by Taylor series expansion of L about $\hat{\theta}$ to yield

$$\frac{1}{p+q+2} (\hat{\theta} - \hat{\theta}_{(i)})^T \mathbf{I}_{\hat{\theta}} (\hat{\theta} - \hat{\theta}_{(i)}).$$

If we replace $\hat{\theta}_{(i)}$ by its one-step approximation

$$\hat{\theta}_{(i)}^1 = \hat{\theta} - \mathbf{I}_{(i)}^{-1} \frac{dl(y_i; \hat{\theta})}{d\theta},$$

we obtain the one-step approximate Cook statistic

$$C_i = \frac{1}{p+q+2} \frac{d l(y_i; \hat{\theta})^T}{d \theta} \mathbf{I}_{(i)}^{-1} \mathbf{I}_{\hat{\theta}} \mathbf{I}_{(i)}^{-1} \frac{d l(y_i; \hat{\theta})}{d \theta}.$$

Here $\mathbf{I}_{(i)}$ is the observed information matrix calculated without observation Y_i . The one-step approximation works provided that the log-likelihood is locally close to quadratic near $\hat{\theta}$, which experience suggests is more nearly the case in the parameterization $\theta = (k, \ln \sigma)$ rather than (k, σ) . Sometimes a $\hat{\theta}_{(i)}^1$ lies outside the parameter space, in which case $\hat{\theta}_{(i)}$ must be found. In our experience the C_i reliably indicate the most influential observations, for which exact Cook statistics can then be found. Exact and approximate Cook statistics can differ quite markedly, but the approximation ranks the largest correctly.

Davison and Tsai (1989) reviewed diagnostics for models with regression and incidental parameters, and showed how to calculate them in standard packages such as GLIM.

5. INFORMATION IN THE TAIL

In modelling statistical extremes it is important to appreciate how estimates typically depend on the few largest observations. This can be made explicit by calculating the influence curves (Hampel *et al.*, 1986) for estimators. For statistical functionals t determined by the implicit equations

$$\int \frac{d l(y; t)}{d \theta} d F(y; \theta) = 0,$$

the theoretical influence curve is

$$IC_{t,F}(y) = \mathbf{i}_{\theta}^{-1} \frac{d l(y; \theta)}{d \theta}. \quad (5.1)$$

Fig. 2 shows influence curves for the maximum likelihood estimators $\hat{k}$ and $\hat{\sigma}$ of the generalized Pareto distribution, for several values of k and $\sigma = 1$. The $IC_{t,F}(y)$ are displayed as functions of $z = \Phi^{-1}\{F(y; \theta)\}$. On this scale the influence curve for the mean, a notoriously sensitive statistic, evaluated at the standard normal distribution, is $IC_{t,F}(y) = y$. The influence on $\hat{k}$ of extreme observations is huge. Fig. 3 shows the $IC_{t,F}(y)$ for the probability weighted moments estimators of k and σ . If anything, these are even more sensitive than the maximum likelihood estimators.

Clearly, outliers can have a big effect when fitting the generalized Pareto distribution. One way to counteract this would be to use more robust estimates. This, however, will lead to a loss of information. To investigate this aspect, we consider the effect of censoring at the top end of the distribution. Suppose that observations arise from the distribution

$$F(y; \sigma, k) = \begin{cases} \{1 - H(y - c)\} \{1 - (1 - ky/\sigma)^{1/k}\} + H(y - c)(1 - kc/\sigma)^{1/k} & k \neq 0 \\ \{1 - H(y - c)\} \{1 - \exp(-y/\sigma)\} + H(y - c)\exp(-c/\sigma) & k = 0, \end{cases}$$

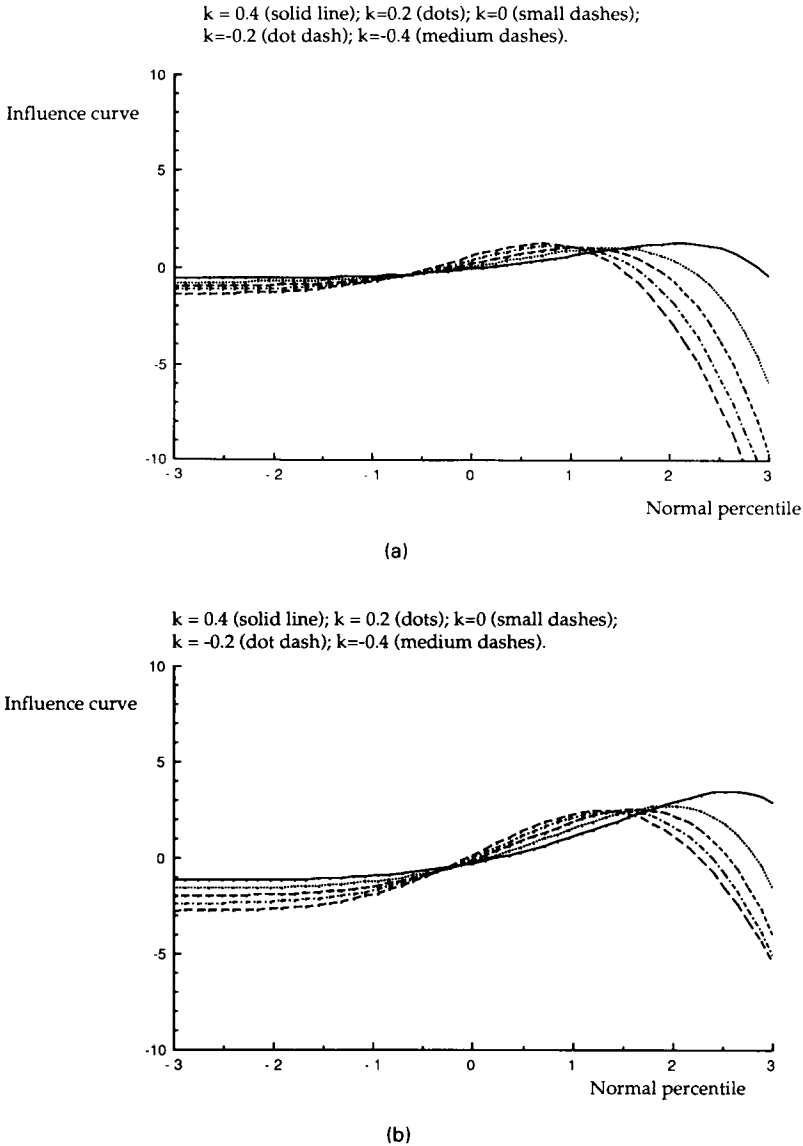


Fig. 2. Influence curves $IC_{t,F}$ for maximum likelihood estimators of k , plotted against normal percentiles: (a) estimator of k ; (b) estimator of σ

where $H(\cdot)$ is the Heaviside function. This distribution is generalized Pareto for $y < c$ but puts an atom of probability of size $p = (1 - kc/\sigma)^{1/k}$ at c if $k \neq 0$ and of size $p = \exp(-c/\sigma)$ there if $k = 0$. When $k < \frac{1}{2}$ the components of the expected information matrix $i_\theta(c)$ for this distribution are readily found from the formulae given in Appendix A, using $w = c/\sigma$. The asymptotic loss of efficiency of estimation of k and σ , $\{|i_\theta(c)| / |i_\theta|\}^{1/2} \times 100\%$, is displayed in Table 1 as a function of the proportion p of the sample censored. The information loss is severe for positive k for p as low as 0.05 and for all values of k considered as p increases from 0.1. In large samples loss of the top 1% of the data may halve the information available. This drastic loss

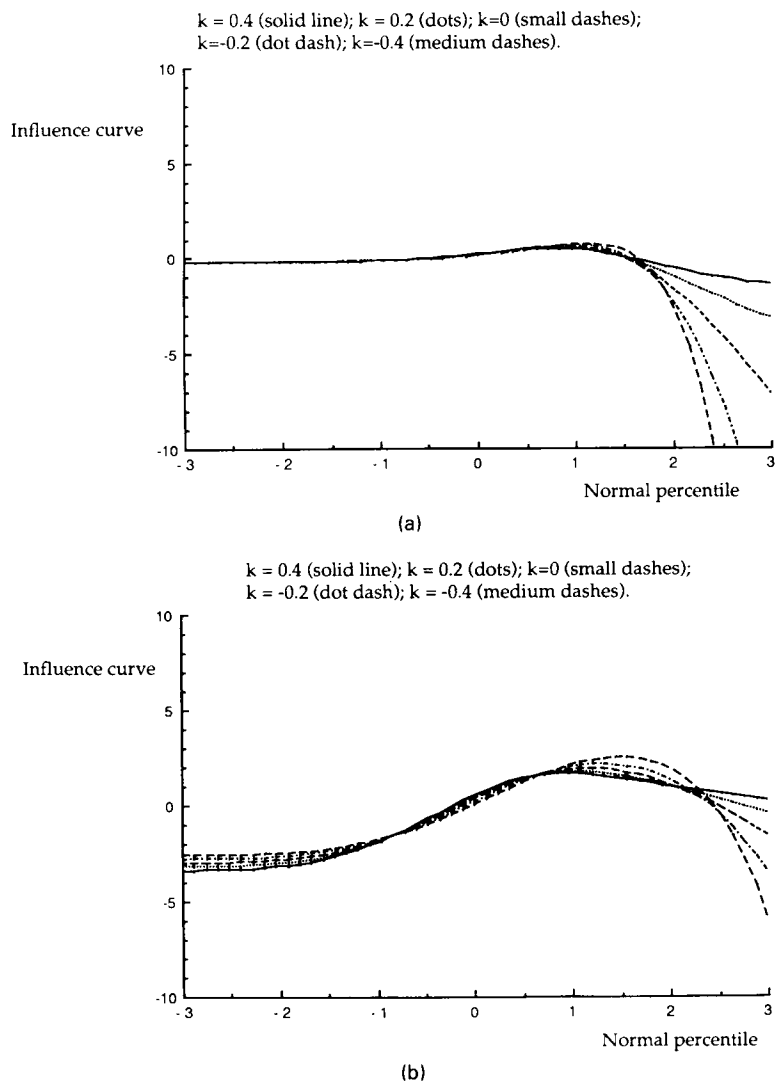


Fig. 3. Influence curves $IC_{t,F}$ for probability weighted moments estimators of k , plotted against normal percentiles: (a) estimator of k ; (b) estimator of σ

TABLE 1
Asymptotic relative efficiency (%) of maximum likelihood estimation from a right-censored generalized Pareto distribution

k	Proportion censored $p \times 100\%$				
	1	5	10	20	50
-0.4	95.4	81.9	69.1	49.9	16.3
-0.2	92.8	76.2	62.2	43.1	13.2
0.0	87.6	67.4	52.9	34.9	9.9
0.2	76.5	53.7	40.1	25.1	6.5
0.4	49.3	30.6	21.5	12.6	3.0

is related to the fact that the uppermost order statistic is a superefficient estimator of the end point when $k > \frac{1}{2}$.

We conclude that in the statistics of extremes the fit of a model is most sensitive to the most informative observations. The statistician is caught in a cleft stick, as use of robust or resistant methods may lead to substantial loss of information. For this reason we advocate the sensitive use of residuals and other diagnostics. Extreme value analysis should never be performed automatically, without the intervention of the data analyst, because failure to spot unusual features of the data could have serious consequences. The comments of Hinkley (1983) on the relation between model fit and robustness are relevant here.

6. APPLICATION IN NUCLEAR SITING POLICY

6.1. *Introduction*

The Chernobyl disaster brought to the public eye the need to assess potential exposures due to long-range atmospheric transport of radioactivity from releases at nuclear installations. For obvious reasons, special interest attaches to higher exposures. This problem has been studied by members of the Nuclear Power Section of the Department of Mechanical Engineering at Imperial College, London, for more than a decade. An early stage of their work was the development of a sophisticated computer model, MESOS. This simulates the atmospheric transport, dispersion and deposition of radionuclides at 100–1000 km distances from notional sources in Western Europe. More recently effort has been spent on summarizing the results of MESOS as a statistical model: MESOSTAT.

MESOS models a hypothetical continuous constant rate release of pollutant from a single source as a plume, which is interpolated between a series of imaginary puffs released every 3 h. The puff trajectories develop as realistically as possible based on 'present weather' observations from meteorological stations and ships throughout Western Europe. Geostrophic winds which advect the puffs are deduced from a series of pressure fields interpolated between three-hourly measurements. The pressure fields, and hence the trajectories, reflect the influence of major orographic features such as the Alps. Each puff is treated as a column of pollutant which expands laterally and evolves vertically according to the local behaviour of the boundary layer at the time of passage. The effects of such dispersion mechanisms as convective turbulence are mimicked. Dispersion of pollutants is combined with the depletion mechanisms of radioactive decay and dry and wet deposition to give estimates of the notional exposures at a receptor fixed relative to the source. Data on three types of exposure are available: time-integrated air concentrations, dry deposition and wet deposition.

The end result is a large number of time series of notional exposures, one of which is displayed in Fig. 4. Characteristic are the skewness of the exposure distribution and clustering together of positive exposures. These are due to the wide diurnal variation in the effect of dispersion mechanisms and the short-term persistence of weather conditions which generate them. Though there probably is seasonal variation in exposure levels, diurnal variation dominates the picture, and we ignore seasonality in our analysis.

Weather data and exposures are available over two periods. The first covers much of 1973, an 'average' year in meteorological terms. The second covers 1976, a more

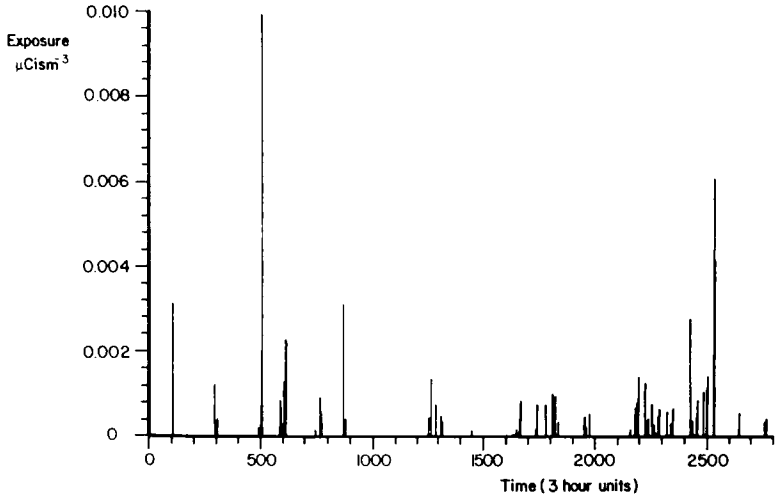


Fig. 4. Time series of exposures to time-integrated air concentration at receptor 800 km north of Mol, due to unit releases of ^{85}Kr every 3 h during 1976

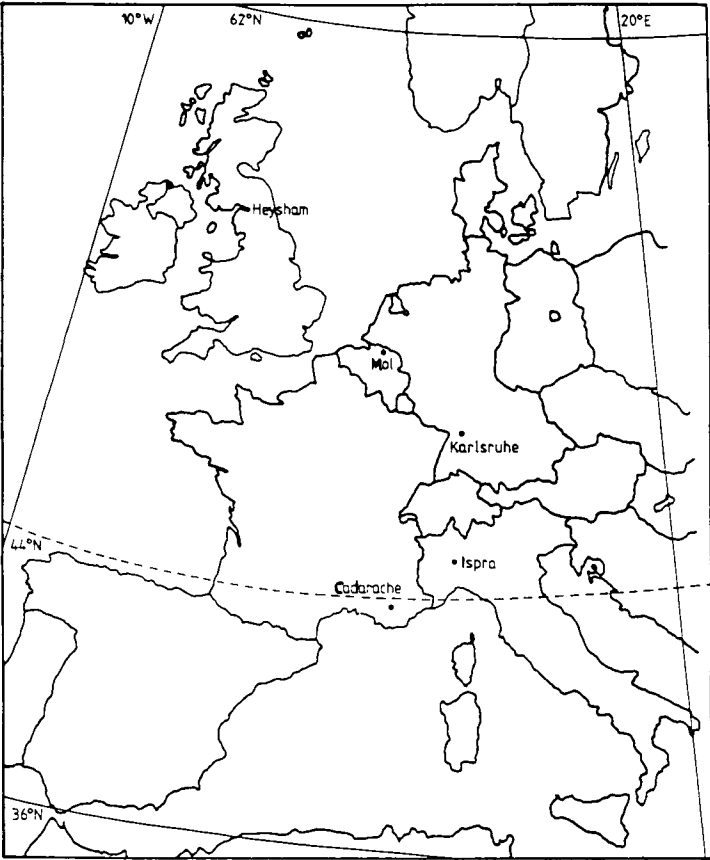


Fig. 5. Area covered by meteorological databases (hypothetical releases from Mol occur in both 1973 and 1976), from ApSimon and Davison (1986): -----, southern limit of the 1973 database at 44°N; —, southern limit of the 1976 database at 36°N

extreme year whose summer was characterized by a number of 'blocking anticyclones' over the British Isles. These result in stagnant airflow and hence potentially higher exposures. For each year there are notional exposures from releases at three sources for each of 12 combinations of nuclide and exposure mode. Fig. 5 shows the source locations. There are 16 receptors between 100 and 1000 km or so away from each source. 12 time series of exposures due to releases every 3 h are available at each receptor. Thus a total of $2 \times 3 \times 16 \times 12 = 1152$ time series of exposures due to releases at three-hour intervals for about a year are available. Time series of exposures for releases of duration longer than 3 h can be obtained by concatenating exposures due to consecutive three-hourly releases. ApSimon *et al.* (1985) describe MESOS and its databases in more detail.

The statistical problem is to condense the exposure data into a concise summary which can be used to assess the possible consequences of the siting policy of one European Economic Community country for the inhabitants of another. Evidently regression modelling is essential. Since our purpose is to illustrate methodology we consider only exposures to time-integrated air contamination. The main features of MESOSTAT are described by ApSimon and Davison (1986).

6.2. Modelling High Exposures

An argument based on simple notions of plume dispersion suggests how exposure levels for a particular release should depend on various covariates. Relevant covariates are the nuclide decay rate λ (s^{-1}), deposition velocity v_d ($m\ s^{-1}$) and washout coefficient Λ_w (s^{-1}), the source-receptor distance d (m) and the duration T (h) of the release. The form of the dependence is

$$\ln(\text{exposure}) = \eta = \beta_0 + \beta_1 d\lambda + \beta_2 dv_d + \beta_3 d\Lambda_w + \beta_4 \ln d + \beta_5 \ln(T/3). \quad (6.1)$$

The parameters have physical interpretations. The mean puff speed and height are $-\beta_1^{-1}$ and β_1/β_2 , the probability of precipitation along a puff trajectory is β_3/β_1 , the rate at which a puff spreads with distance is $-\beta_4$, and the rate at which a puff spreads with release duration, measured in three-hour units, is $-\beta_5$.

A single threshold cannot be applied for all receptors and radionuclides. Exposure levels depend heavily on source-receptor distance and radionuclide parameters, and the model is to be used to predict exposures for source-receptor combinations not included in the original data. This implies that the threshold used for each time series must depend on the covariates. Accordingly equation (6.1) was fitted by ordinary least squares to the log-quantiles of the exposures, in the range 0.9–0.99. Only β_0 changed substantially as the quantiles become more extreme. Estimates of $\beta_1, \dots, \beta_5$ obtained for the 0.9 quantile were then regarded as fixed and the value of β_0 varied to prescribe three thresholds $u_0 < u_1 < u_2$ for each time series.

Only one of the six available time-integrated air concentration time series for each source-receptor combination was included in the analysis, so that the assumption of independence between series should not be violated too grossly. After ignoring data below the lowest threshold u_0 for each of the 96 time series, 3524 observations were available. These were split into two sets of equal size, one of which was put aside for cross-validation. The basic model used is that the time series are independent and that, for each, excesses over its threshold are independent identically distributed generalized Pareto random variables with parameters k and $\sigma = \exp \eta$.

TABLE 2
Reductions in -2 log-likelihood due to fitting covariates to time-integrated air concentration data in the order given

Covariate	Threshold		
	u_0	u_1	u_2
$\ln(T/3)$	142.0	44.4	22.4
$\ln d$	1438.0	769.2	320.0
$d\lambda$	432.0	274.6	158.0
$d\Lambda_w$	66.0	21.6	6.4
dv_d	16.0	5.8	0.8
No. of cases	1762	751	280

TABLE 3
Parameter estimates for exposures to time-integrated air concentrations, for different thresholds u

Parameter	Threshold					
	Estimate	u_0 Standard error	Estimate	u_1 Standard error	Estimate	u_2 Standard error
β_0	23.4	0.50	25.0	0.75	25.1	1.18
β_1	-0.14	4.9×10^{-3}	-0.14	6.7×10^{-3}	-0.14	9.5×10^{-3}
β_2	-2.30×10^{-4}	5.5×10^{-5}	-1.80×10^{-4}	7.68×10^{-5}	-0.89×10^{-4}	1.20×10^{-4}
β_3	-9.89×10^{-3}	3.14×10^{-3}	-8.69×10^{-3}	4.58×10^{-3}	-1.11×10^{-2}	7.32×10^{-3}
β_4	-1.23	4.00×10^{-2}	-1.32	5.99×10^{-2}	-1.29	9.42×10^{-2}
β_5	-0.72	3.70×10^{-2}	-0.84	4.9×10^{-2}	-1.07	7.85×10^{-2}
k	-0.23	3.10×10^{-2}	-0.12	4.09×10^{-2}	-0.05	6.29×10^{-2}

Tables 2 and 3 summarize the fit of the model. At each threshold all covariates except dv_d give significant reductions in minus log-likelihood. The biggest effects at all thresholds are of spread with distance d and of $d\lambda$. At higher thresholds the effect $\hat{\beta}_5$ of release duration T becomes more marked and the tail becomes more nearly exponential. The parameter estimates suggest that for high exposures the mean puff speed is 7 m s^{-1} , the mean puff height is 600 m and the proportion of travel time over which rain is experienced is about 0.07. The puff height is a little low, but otherwise these values are in broad agreement with physical reasoning. Comparison of the value of $\hat{\beta}_4$ with its value for all exposures suggests that plumes broaden more rapidly with distance for high exposures than for all exposures. This agrees with physical intuition.

Confidence intervals for the true k based on the profile likelihood, on observed and on expected information agree closely for each threshold. A hypothesis of interest is that different dispersion conditions apply to releases over periods of 12 h or less and those over periods of 1 day or more, because of the effect of the diurnal mixing cycle. For threshold u_0 a likelihood ratio test gives evidence at between the 5% and 1% levels that there are different values of k for releases of duration 1 day or longer and those of duration 12 h or shorter. The value of $\hat{k}$ for exposures due to releases over periods of 12 h or less is -0.25 , but that for exposures due to releases of longer duration is -0.08 . This confirms that exposures due to releases of shorter duration

have heavier tails. A similar effect is found for the effect of source–receptor distance d on exposures: tails are heavier closer to the source.

An Anderson–Darling test using the parameter estimates for u_0 was applied to the half of the data initially put aside. The observed value of the statistic is 0.084, not nearly significant even at the 15% level. Residual plots show no big anomalies, though the variability of the residuals increases somewhat with the linear predictor η , presumably due to the heterogeneity of values of k . Plots of approximate Cook statistics reveal no unusually influential observations, which is hardly surprising when $N=1762$ and the covariate matrix is well balanced. At higher thresholds the fit is generally better, and in particular evidence for heterogeneity of the values of k vanishes.

Our discussion has ignored the clustering of exposures and has treated exposures in the same time series as independent. Some justification for this is that in this application attention is directed at the marginal distribution of exposures. However, the clustering of exposures does affect return values for several unit releases in succession. A rather simplistic model for clustering which takes this into account is given by ApSimon and Davison (1986), and more sophisticated models for clustering and serial dependence are described below.

Failure of the independence assumption would mean that the standard errors quoted above are too small. Randomization tests of the hypotheses that cluster means differ or that there is correlation among successive exposures within clusters give no evidence of these effects, so the standard errors are probably an accurate reflection of the statistical uncertainty in parameter estimates based on these data. However, statistical uncertainty here is dwarfed by the modelling uncertainties—possibly of an order of magnitude—inherent in data generated by a complex transport model like MESOS. This puts concerns about independence assumptions into perspective.

The analysis described above leads to a concise and physically interpretable model for extreme exposures, applicable for a wide range of radionuclides at distances of 100–1000 km from a possible source. Whatever the shortcomings of the computer model on which a statistical model must be based, regression analysis is essential if predictions are to be made for sources, receptors and radionuclides not included in the original study.

7. EXTREME VALUES IN ENVIRONMENTAL TIME SERIES

Environmental time series such as river levels or air pollution measurements typically exhibit two features which are not captured by the model of Section 2, namely serial dependence and seasonality, even when there are no other regressors as in the model of Section 6.

The main contribution of the probabilistic theory of extremes has been a very detailed study of extreme values in stationary stochastic processes; see the monograph of Leadbetter *et al.* (1983) and the more recent review papers of Lindgren and Rootzén (1987) and Leadbetter and Rootzén (1988). Most of the results of this theory are of the form that, under suitable mixing conditions, the high level excesses behave asymptotically as a clustered Poisson process. This suggests that our basic model, of a Poisson process of exceedance times with conditionally independent excesses, remains valid for the *cluster* maxima of a wide class of stationary processes. The use of the generalized Pareto distribution for the peak excess in each cluster may again

be justified by arguments similar to those in Section 2. A key parameter in such models is the *extremal index*, θ , defined by the relation

$$\text{pr}(M_n \leq x) \approx F^{n\theta}(x)$$

where M_n is the maximum of n consecutive observations and F is the marginal distribution function of the process. The extremal index lies between zero and one ($\theta = 1$ is the case of no clustering), and its reciprocal is the limiting mean cluster size. More detailed studies of the extremal index are in the papers of O'Brien (1974), Leadbetter (1983), Hsing *et al.* (1988) and Smith (1988a).

The extension of this theory to non-stationary series has been considered (Hüsler, 1986; Smith, 1988b), but the results are too broad to suggest useful statistical procedures. Models for seasonal series will be considered in Sections 8 and 9.

For the remainder of this section, we discuss some methods for identifying clusters. There are two approaches, parametric and nonparametric. The parametric approach is to assume some definite stochastic model for the point process, estimating the parameters. Nonparametric methods are more *ad hoc* but avoid the specification of a precise model.

Among possible parametric models are the well-known Neyman–Scott and Bartlett–Lewis processes, Markov chain models (Stern and Coe, 1984) and doubly stochastic processes. A specific example of the last was proposed by Smith (1984); this raises some general issues concerning the use of parametric models in this context.

In this model, the unobserved state process $\Lambda(t)$ is taken to be a two-state Markov chain with jump rates λ (from zero to one) and μ (from one to zero). Conditionally on Λ , the exceedance times form a Poisson process with rate $\phi\Lambda(t)$. This may be thought of as representing the background state of the environment, such as 'stormy' or 'calm'. It may be checked that the interevent times are independent with common density

$$f(z) = \frac{\phi}{\alpha_1 - \alpha_2} \{(\mu + \phi - \alpha_2)\exp(-\alpha_1 z) - (\mu + \phi - \alpha_1)\exp(-\alpha_2 z)\}, \quad z > 0, \quad (7.1)$$

where α_1 and α_2 are the roots (positive and distinct) of the quadratic equation

$$\alpha^2 - (\lambda + \mu + \phi)\alpha + \lambda\phi = 0.$$

These formulae follow from Grandell (1976), pp. 40–44. Maximum likelihood estimation of the parameters λ , μ and ϕ is easily carried out numerically.

We also need some means of identifying clusters. Intuitively, T_{i-1} and T_i belong to the same cluster if $\Lambda(t) = 1$ for $T_{i-1} < t < T_i$. Under this definition, the mean number of clusters per unit time is $\lambda\mu\phi / \{(\lambda + \mu)(\mu + \phi)\}$. In practice, we assign consecutive values T_{i-1} and T_i to the same cluster if $T_i - T_{i-1} < z^*$, for some z^* . The rule that we adopt is to determine z^* so that the mean number of clusters per unit time equals the theoretical value obtained from the doubly stochastic model. This leads to the formula

$$\int_{z^*}^{\infty} f(z) dz = \frac{\lambda\mu\phi}{(\lambda + \mu)(\mu + \phi)} \int_0^{\infty} zf(z) dz = \frac{\mu}{\mu + \phi}. \quad (7.2)$$

The idea then is to use this rule to identify clusters and hence cluster maxima, and then to fit the generalized Pareto distribution to the cluster maxima. The extremal index is $\theta = \mu/(\mu + \phi)$.

This approach, of cluster identification via a parametric model, may be compared with a nonparametric approach in which essentially the same rule is used for identifying clusters, but with z^* determined empirically and the extremal index estimated as the reciprocal of mean cluster size. In several cases which have been studied, the results have been insensitive to the precise value of z^* , provided that something intuitively reasonable is used. For example Tawn (1988), analysing sea-level data from the east coast of England (where the tidal effect is relatively mild), found that values of z^* between 15 h and 30 h gave comparable results, and it was only for values well outside this range that serious distortions occurred. Smith's (1989) analysis of ozone data at Houston used $z^* = 24$ h or $z^* = 72$ h, with little difference between the results in the two cases. In cases such as these, little seems to be gained by fitting a parametric model.

8. MODELS FOR SEASONALITY

For handling seasonal data we identify two basic approaches.

- (a) It is possible to 'prewhiten' the series by removing known seasonal components before carrying out the extreme value analysis. Examples include the 'joint probabilities method' (Pugh and Vassie, 1980; Tawn and Vassie, 1989), in which sea-level data are decomposed into tidal and surge components, and Smith (1984). The potential disadvantage of this is that seasonal effects identified from the central portion of the data may not be valid in the tails, so the method seems best confined to cases where the physical origin of the seasonal component is well understood.
- (b) The principal alternative is a 'separate seasons' approach in which the year is broken up into a finite number of seasons, separate models being fitted within each season. This approach is inherent in Carter and Challenor (1981), who fitted separate extreme value distributions to each month's data, and the analysis of Texas ozone data by Smith (1989).

The present section describes one version of the prewhitening model, while the next gives an example of the separate seasons approach.

The model considered here assumes that the observed series $X(t)$ is of the form

$$X(t) = M + A \cos(\omega t + \eta) + Z(t) \quad (8.1)$$

where $Z(t)$ is a stationary series and ω , M , A and η are constants. This is appropriate when the only influence of seasonality is in the form of a sinusoidal, additive seasonal factor. It is assumed that ω is known and M , A and η may be taken to be either known or unknown, in the latter case estimated from the whole series. The original series may thus be filtered to obtain $Z(t)$, for which we now fit the generalized Pareto distribution to all peaks over a fixed threshold u . In this section we depart slightly from the previous section by assuming the *clusters* to form a Poisson process of rate λ^* .

The chief complication in this approach lies in calculating return levels. Let $H(q)$ denote the probability that all values of the series X in a single year are below the level q . For $q > u + M + A$, a standard Poisson process calculation leads to the formula

$$H(q) = \exp \left[- \frac{2\lambda^*}{\omega} \int_0^\pi \left\{ 1 - \frac{k(q - M - u - A \cos t)}{\sigma} \right\}^{1/k} dt \right] \quad (8.2)$$

where k and σ are the generalized Pareto distribution parameters and λ^* is the mean number of clusters per unit time. When $k = 0$, the integral in equation (8.2) simplifies to

$$\int_0^\pi \exp\{-(q - M - u - A \cos t)/\sigma\} dt = \exp\{-(q - M - u)/\sigma\} \pi I_0(A/\sigma)$$

where I_0 is the modified Bessel function of order zero (Abramowitz and Stegun (1964), equation 9.6.16). The n -year return level, for large n , is then obtained by inverting this expression to give

$$q_n \approx M + u + \sigma \log \left\{ \frac{2\pi n \lambda^* I_0(A/\sigma)}{\omega} \right\}. \quad (8.3)$$

Approximate standard errors may be found by the delta method; in this connection it is worth noting that $I'_0 = I_1$ (Abramowitz and Stegun (1964), equation 9.6.27).

As an example of this approach, we consider 7 years of wave height data at Seven Stones in the English Channel. Plots of the original series showed a strong sinusoidal seasonal effect but one that affected both the mean and variance of the data, i.e. it did not appear to be an additive model. However, when the same plots were made of the logged series and an additive seasonal component removed, standard time series tests applied to the residuals revealed no evidence of seasonality. Consequently model (8.1) was assumed with $X(t)$ the logarithms of the original series in metres. The parameters M and A were estimated from the whole series as $M = 0.710$ and $A = 0.402$. In the following analysis these are treated as known constants. The data were collected three hourly, so $\omega = 2\pi/2922$, the denominator being the mean number of measurements per year.

Four thresholds were chosen, 1.0, 1.1, 1.2 and 1.3. These four thresholds correspond to mean numbers of peaks per year between 1.5 and 12. In practice it is often recommended (for example in the *Flood Studies Report* (Natural Environment Research Council, 1975)) that the threshold be chosen to fix a value between one and five peaks per year, so the range of thresholds used here covers a somewhat larger range than that advice would allow. First all *exceedances* over the threshold were noted and the model (7.1) fitted to the interexceedance times. This enabled λ , μ and ϕ to be estimated, and hence $\lambda^* = \lambda\phi/(\mu + \phi)$ and the extremal index $\theta = \mu/(\mu + \phi)$. Having thus identified clusters, it was possible to identify *peaks* within clusters and hence the *excesses* of the peaks over the threshold.

Likelihood ratio tests of the hypothesis $k = 0$ suggested that the generalized Pareto distribution for the excesses could be reduced to the exponential distribution ($k = 0$)—for instance, for the threshold $u = 1.1$, the deviance statistic (nominally χ^2_1) is 0.92. Similar results were obtained for the other thresholds. Henceforth we assume $k = 0$. Parameter values under this model are shown in Table 4, with standard errors in parentheses.

The estimated 50-year return levels for the log(wave heights) are consistent with each other and also with estimates produced by other studies of the same data (Carter and Challenor, 1981; Challenor, 1982), whose estimates ranged from 2.52 to 2.73 with no standard errors quoted. This seems satisfactory, but even so the standard

TABLE 4
Seven Stones data: estimates for four thresholds†

Threshold	1.0	1.1	1.2	1.3
No. of exceedances	295	140	74	36
No. of peaks	81	43	27	11
ϕ	0.46 (0.04)	0.52 (0.06)	0.47 (0.08)	0.50 (0.11)
λ	0.0067 (0.0008)	0.0036 (0.0006)	0.0025 (0.0005)	0.0009 (0.0003)
μ	0.18 (0.02)	0.23 (0.04)	0.27 (0.06)	0.22 (0.07)
θ	0.28 (0.03)	0.31 (0.04)	0.37 (0.06)	0.31 (0.08)
σ	0.15 (0.02)	0.15 (0.02)	0.12 (0.02)	0.12 (0.04)
q_{50}	2.75 (0.05)	2.77 (0.07)	2.72 (0.08)	2.68 (0.12)

† The units of ϕ^{-1} , μ^{-1} and λ^{-1} are the time interval between observations, i.e. 3 h.

errors are large when their practical implications are considered. For instance, even with $u = 1.0$ the approximate 95% confidence interval for q_{50} is (2.65, 2.85), or (14.2, 17.3) on the original scale. An addition of 3 m is quite significant when the costs of building an oil platform are considered!

It is possible to perform a number of sensitivity analyses of the various assumptions made in this analysis. For example, model (8.1) assumes that the seasonal component is additive. Our analysis depends on this assumption remaining valid for the extreme values of the process. Standard time series tests for seasonality are not adequate for this. However, it is possible to devise tests for seasonality based purely on the exceedances over a high threshold. As an example of this, Smith (1984) defined a *total excess statistic* $\sum(Z(t) - u)_+$ calculated for each month of the series. In other words, the sum of all excesses over the threshold is calculated. In the present example this procedure revealed no evidence of any residual seasonality in the excess values. Other tests include tests of independence for the successive excess values and sensitivity analyses of assumptions such as $k = 0$. In the present example, the latter analysis did reveal cause for concern, but we do not pursue this here as a more detailed study of this kind of analysis is given in the next section.

The analysis here is adapted from Smith (1984); we thank Jonathan Tawn for pointing out a numerical error in the return values quoted there.

9. HIGH LEVEL EXCEEDANCES OF THE RIVER NIDD

This section of this paper presents an example which illustrates the difficulties which sometimes arise in using these methods to estimate extreme quantiles.

The data consist of 154 exceedances of the threshold level $65 \text{ m}^3 \text{ s}^{-1}$ by the River Nidd at Hunsingore Weir from 1934 to 1969 (35 years). The source is pp. 235–236 of the *Flood Studies Report* (Natural Environment Research Council, 1975), volume 4. Previous analyses are in volume 1 of the *Flood Studies Report* and Hosking *et al.* (1985), based on the annual maxima, and by Hosking and Wallis (1987) using a threshold approach. The data already satisfy some independence criteria (Natural

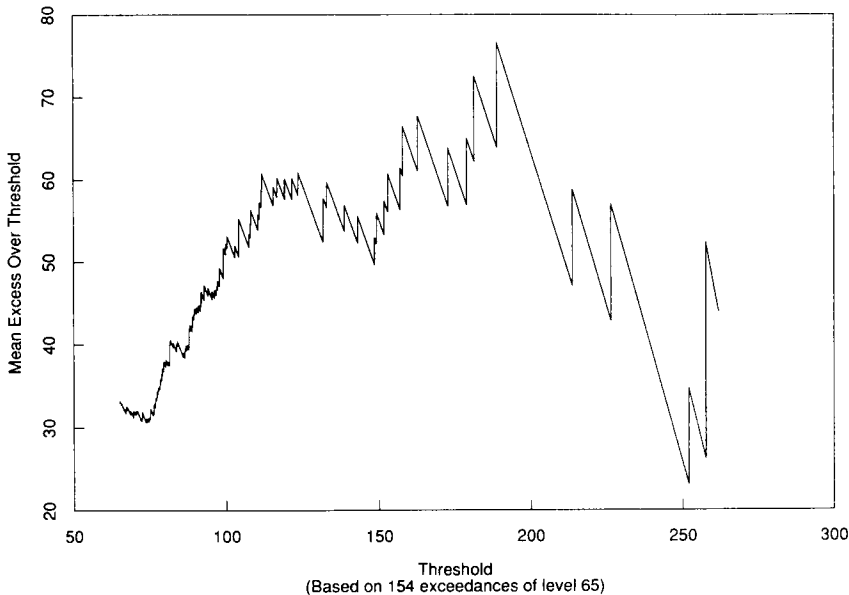


Fig. 6. Mean residual life plot for the Nidd data

Environment Research Council (1975), volume 4, p. 13) so for most of the analysis we do not distinguish between exceedances and peaks. A sensitivity analysis of this assumption is made later.

Fig. 6 shows a mean residual life plot for these data. The plot shows a rapid increase up to around the threshold 110, then levels off. The very large fluctuations at the right-hand side are the result of sampling variation in the top few order statistics and are not significant. The levelling off near 110, however, does appear to be significant, and suggests there may be some sort of mixture of two populations. This kind of feature has been observed in several other environmental series that we have examined.

Several thresholds were tried for the analysis, with results summarized in Table 5. These show maximum likelihood estimates for the three parameters, with standard errors, plus point estimates and standard errors for the 25-, 50- and 100-year return levels. The results confirm the picture of Fig. 6, in that there seems to be a substantial change in the value of both k and the return level estimates around threshold 100, though there is high variability in all the estimates. Hosking and Wallis (1987) gave estimates using probability weighted moments.

Figs 7 and 8 show quantile-quantile plots for the fitted models at thresholds 70 and 100. The message of Fig. 7 is that the model fits the data except in the largest two order statistics. In other contexts it might be appropriate simply to drop the largest two observations, but this does not seem appropriate here as our main interest is in extrapolation. In contrast, Fig. 8 shows no such difficulty.

This impression is partly confirmed by goodness-of-fit statistics. The renormalized Kolmogorov-Smirnov and Anderson-Darling statistics are given in Table 6. A comparison with the 5% critical points for testing an exponential distribution with unknown mean (Pearson and Hartley, 1976) suggests that the values are not significant for any of the thresholds considered, though it is noticeable that both test statistics

TABLE 5

Summary of parameter estimates for maximum likelihood fits of the generalized Pareto distribution to the Nidd data†

Threshold	No. of exceedances	λ	σ	k	q_{25}	q_{50}	q_{100}
140	18	0.51 (0.12)	66 (20)	0.24 (0.26)	266 (24)	289 (31)	309 (44)
130	22	0.63 (0.13)	59 (21)	0.14 (0.29)	265 (27)	292 (39)	316 (56)
120	24	0.69 (0.14)	72 (24)	0.25 (0.22)	266 (23)	288 (30)	307 (39)
110	30	0.89 (0.16)	56 (17)	0.07 (0.24)	267 (32)	298 (47)	327 (67)
100	39	1.11 (0.18)	51 (14)	0.00 (0.21)	269 (37)	305 (56)	340 (80)
90	57	1.63 (0.22)	34 (8.1)	-0.24 (0.20)	290 (61)	351 (101)	423 (158)
80	86	2.46 (0.27)	25 (4.9)	-0.34 (0.16)	308 (73)	389 (122)	492 (196)
70	137	3.94 (0.34)	22 (3.0)	-0.32 (0.11)	298 (60)	372 (96)	465 (148)

†Standard errors are given in parentheses.

TABLE 6

Goodness-of-fit statistics for maximum likelihood fits of the generalized Pareto distribution to the Nidd data†

Test statistic (renormalized)	Threshold								5% critical points
	140	130	120	110	100	90	80	70	
Kolmogorov-Smirnov	0.60	0.54	0.52	0.55	0.63	0.56	0.50	0.88	1.09
Anderson-Darling	0.42	0.36	0.27	0.51	0.31	0.35	0.37	0.87	1.34

†Critical points are taken from exponential tables (Pearson and Hartley, 1976), so are overestimates of the true critical values.

increase sharply at the lowest threshold (70). Quoting these tables in this context is suspect since the exponential distribution is only a submodel of the generalized Pareto distribution, so the true critical points are smaller, but they are the nearest published tables to those that we want.

We now consider the use of these results to obtain confidence intervals for return levels. Relatively short return periods of 25, 50 and 100 years are considered because even these illustrate substantial difficulties; hydrologists are sometimes interested in return periods of 250 or 500 years or even more, but in such cases the difficulties would be even worse.

The standard errors quoted in Table 5 are so large that the Wald 95% confidence interval (parameter estimate $\pm 1.96 \times$ standard error) is unlikely to produce good results, even after a logarithmic transformation of the parameter. Therefore, we construct interval estimates based on the log-profile-likelihood of the return level. To achieve this, we reparameterize (σ, k, λ) as (q_n, k, λ) via equation (2.3), where

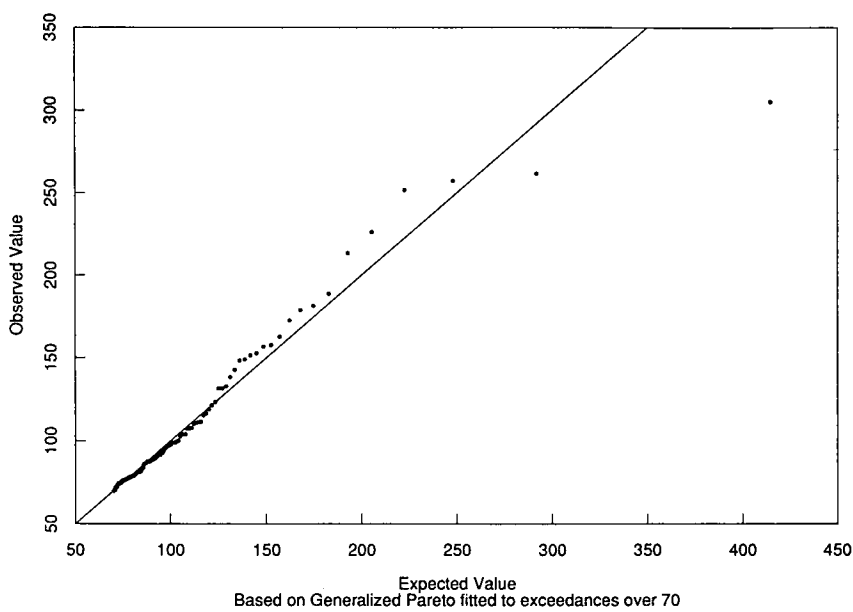


Fig. 7. Quantile-quantile plot for the Nidd data, threshold $u = 70$

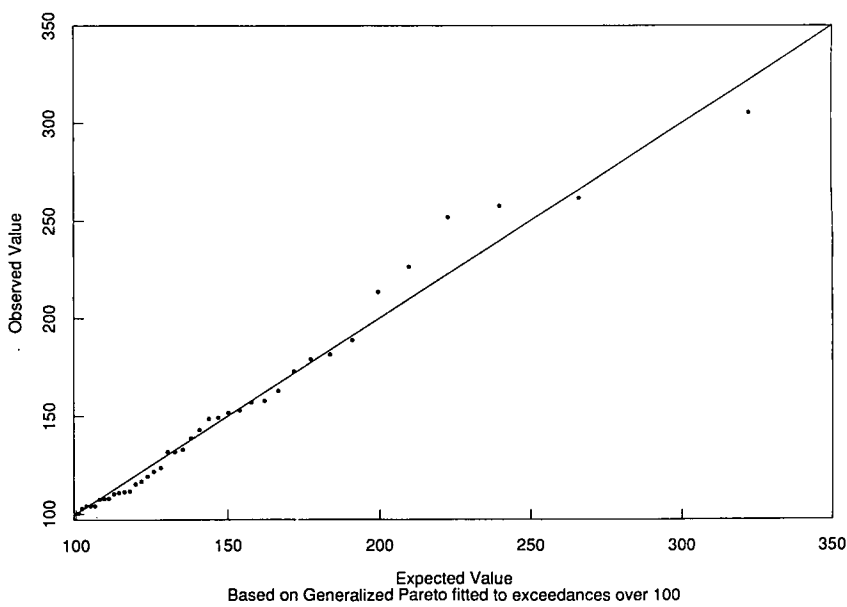


Fig. 8. Quantile-quantile plot for the Nidd data, threshold $u = 100$

q_n is the desired return value. By maximizing the log-likelihood with respect to k and λ , we obtain a profile log-likelihood for q_n and hence, via the usual χ^2_1 approximation, an approximate 95% confidence interval.

Even this procedure is open to the objection that it is based on asymptotic theory and, with small sample sizes for what is being estimated, this may not be applicable.

TABLE 7
Likelihood-based and Bayesian 95% confidence intervals for return levels for the Nidd data

<i>Analysis</i>	<i>Point estimate</i>	<i>Standard error</i>	<i>Profile likelihood</i>	<i>Bayesian</i>
<i>25-year return level</i>				
<i>u</i> = 100	269	37	(220, 465)	(205, 580)
<i>u</i> = 120	266	23	(225, 390)	(210, 665)
<i>u</i> = 100, declustered	268	34	(220, 450)	(205, 605)
<i>u</i> = 100, top 2 censored	300	67	(225, 715)	(200, 565)
<i>50-year return level</i>				
<i>u</i> = 100	305	56	(240, 650)	(210, 775)
<i>u</i> = 120	280	30	(245, 495)	(215, 850)
<i>u</i> = 100, declustered	300	50	(245, 615)	(215, 805)
<i>u</i> = 100, top 2 censored	357	110	(250, > 1000)	(210, 765)
<i>100-year return level</i>				
<i>u</i> = 100	340	80	(260, 925)	(220, 935)
<i>u</i> = 120	307	39	(265, 630)	(225, 945)
<i>u</i> = 100, declustered	332	71	(260, 850)	(225, 945)
<i>u</i> = 100, top 2 censored	421	170	(270, > 1000)	(220, 930)

This suggests looking for a non-asymptotic method. There is no exact frequentist procedure in this situation but it is possible to perform a Bayesian analysis. One motivation for this is that, in a previous example similar in some respects to the present one, Smith and Naylor (1987) found that the Bayesian coverage intervals were much shorter than those based on the profile likelihood and argued that this was a practical reason for preferring the Bayesian analysis. Another motivation is the connection between Bayesian methods and the Cox and Reid (1987) procedure for constructing a modified profile likelihood based on an orthogonal parameterization (see the contributions of T. J. Sweeting and R. L. Smith to the discussion of Cox and Reid). Indeed the Cox-Reid procedure may itself be an alternative for this problem but we have not succeeded in solving the necessary orthogonalization equations.

Table 7 quotes 95% coverage intervals for the return levels, based on both profile likelihood and Bayesian analysis. The Bayesian intervals are based on uniform prior distributions over (200, 1000) for q_n , (0.5, 1.7) for λ , except that the interval used was (0.2, 1.2) for threshold 120, and (−0.5, 0.5) for k . These prior distributions are intended only for illustration, and comparison with the profile likelihood procedure, and do not pretend to represent actual prior information. The computational method used Numerical Algorithms Group (1983) routine D01DAF to integrate out k and λ at each $q_n=200, 205, 210, \dots, 1000$, followed by summation over the grid of values of q_n .

These intervals are constructed on the basis of four different analyses of the data—those based on thresholds 100 and 120, a ‘declustered data’ analysis which will be described in a moment and an analysis in which the largest two observations are treated as censored. As might be expected from Figs 7 and 8, this has the effect of increasing the estimates, thus reflecting the influence which these two observations have on the return level estimates. In contrast, the measure of influence via Cook statistics does not show any observations to be highly influential. A more useful diagnostic would assess the effect of data perturbation on q_n directly, perhaps via local influence (Cook, 1986).

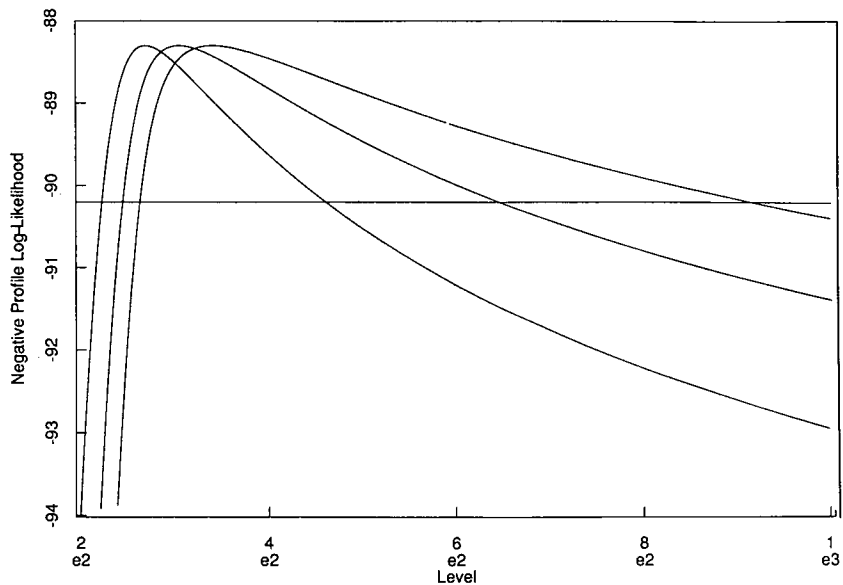


Fig. 9. Profile log-likelihood for 25-, 50- and 100-year return levels for the Nidd data (the horizontal line gives confidence intervals based on an asymptotic χ^2_1 distribution)

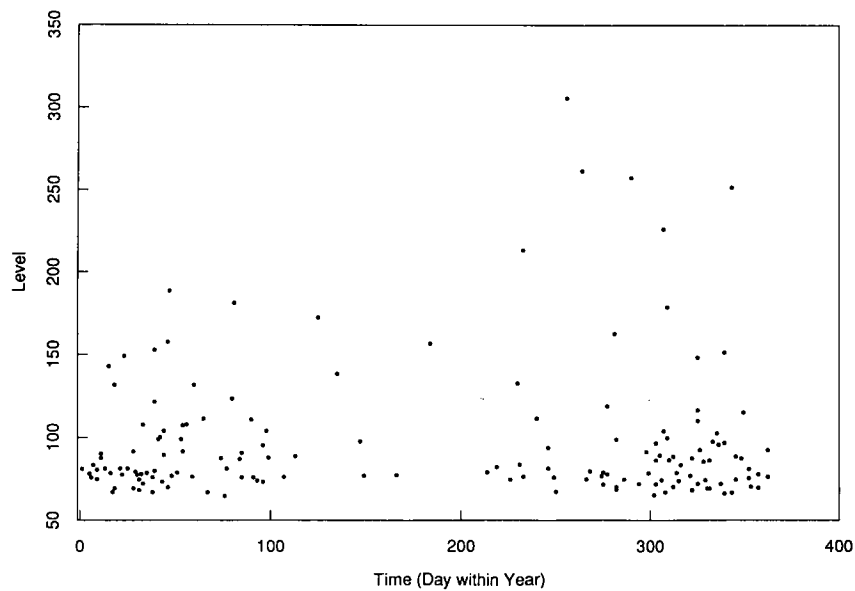


Fig. 10. Plot of exceedances of the Nidd data over threshold 65 against day of year

A plot of the profile log-likelihoods for the analysis based on threshold 100 is shown in Fig. 9. The horizontal line represents the χ^2_1 cut-off value for the confidence intervals. It can be seen that all the intervals quoted in Table 7 are very wide. In contrast with the example studied by Smith and Naylor (1987), the Bayesian intervals are not necessarily shorter than the profile likelihood intervals. The one feature that does

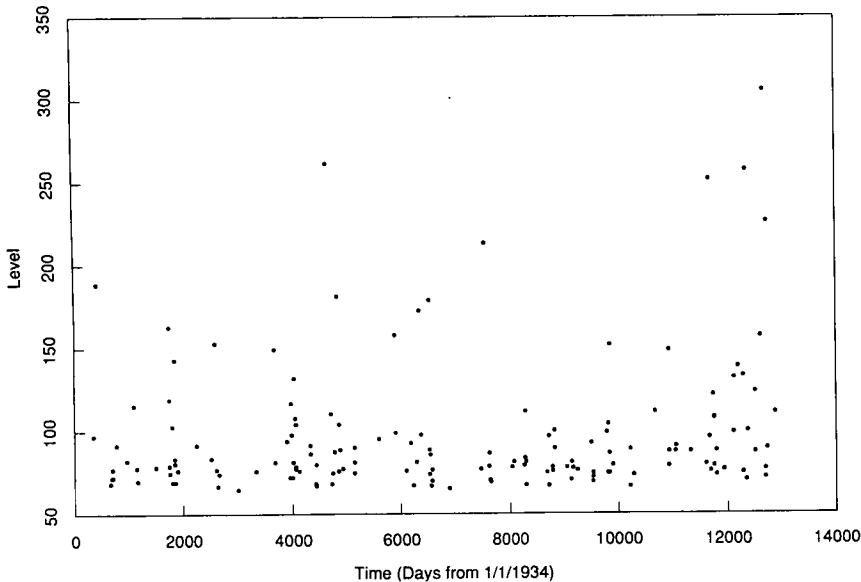


Fig. 11. Plot of exceedances of the Nidd data over threshold 65 against time of occurrence

commend the Bayesian intervals is that they seem to be more stable over the different analyses, though it is not clear to what extent this is the result of using the same prior distribution in the different cases.

A more detailed consideration of this data set suggests that we should look for trends and seasonal effects, as well as studying the effect of clustering. Plots of the data against time of year in Fig. 10 and against overall time in Fig. 11 suggest some effects—from Fig. 10, it appears that most of the large values are in the second half of the calendar year, while in Fig. 11 it can be seen that many of the largest values occurred towards the end of the data set. These effects are separate from the most obvious seasonal effect, that there are far fewer high excesses in summer, something which would not in itself invalidate the earlier analysis.

The foregoing analysis has taken no account of clustering. As a check on this, the analysis was repeated using cluster peaks as identified by the method of Section 7, with $z^* = 5$ days. This had the effect of reducing the total number of peaks from 154 to 116. However, when applied to estimation of return levels there was hardly any effect, as Table 7 shows.

In a seasonal analysis, the main issue is whether the value of k is seasonally dependent. This was examined using the 'separate seasons' approach mentioned in Section 8. In this case, it was considered that the small number of peaks over threshold 100 made it impractical to use more than two seasons, so separate analyses were performed for January–June and July–December. This would be expected to pick up the discrepancies observed from Fig. 10. The threshold was fixed at 100 throughout the present analysis.

Splitting the data in this way has a surprising effect: the generalized Pareto parameter estimates are $\hat{\sigma} = 56$ (standard error 20) and $\hat{k} = 0.58$ (0.30) for January–June, $\hat{\sigma} = 110$ (38) and $\hat{k} = 0.45$ (0.29) for July–December. This is in very sharp contrast with the values in Table 5. As expected, this also influences the estimated return values.

For example, 25-, 50- and 100-year estimates based solely on the data in January–June are 175 (standard error 9), 183 (9) and 189 (12), while those based on July–December are 266 (24), 289 (30) and 307 (40). These are much smaller (and with smaller standard errors) than the estimates quoted earlier based on the full data.

The effect is deceptive, however. Profile likelihood confidence intervals are (160, 225), (170, 270) and (175, 320) based on January–June, and (220, 440), (245, 625), (265, 895) based on July–December. The last three intervals are not very different from those quoted in Table 7. Thus the conclusion is that restriction to July–December does not much affect the interval estimates even though it has a dramatic effect on the parameters—providing further support for the use of profile likelihood in place of Wald-type intervals based on standard errors.

Our conclusion is that there is a pronounced seasonal effect, with few excesses during the summer and all the highest excesses occurring during the second half of the year. However, the effect of this on estimates of return levels is minimal.

A final effect which might be studied is that of trend, especially since Fig. 11 shows that most of the highest excesses occurred towards the end of the period under study. It is possible to consider the effect of (say) an additive linear trend as follows. Under our basic model in which exceedances of a threshold u are Poisson distributed with intensity λ (per year) and the excesses over u are independent generalized Pareto distributed, the probability that a level $y > u$ is not exceeded during the year is

$$\exp \left\{ -\lambda \left(1 - k \frac{y-u}{\sigma} \right)^{1/k} \right\}.$$

This is of generalized extreme value form (1.1) with

$$\lambda = \{1 - k(u - \mu)/\psi\}^{1/k}, \quad \sigma = \psi - k(u - \mu). \quad (9.1)$$

An additive trend corresponds to μ being time dependent. If we write, for example, $\mu = \alpha + \beta t$ for a linear trend in time t , transform to (σ, k, λ) via equations (9.1) and substitute in the likelihood for (σ, k, λ) , we obtain a likelihood function for the model with trend. This is a variant on the method of Smith (1989).

In the present case, looking for trends had an effect which might be anticipated from Fig. 11. Fitting to the whole data set produced a positive trend which was not quite significant but still sufficiently great to have a substantial effect on estimates of extreme quantiles. However, refitting just to the data up to 1964 (omitting the last few years) produced a smaller trend in the opposite direction. Once again, it appears that a few large observations are calling the tune.

In summary:

- (a) *clustering* is present but has very little effect on return level estimates;
- (b) with regard to *seasonality*, there is clear evidence of an overall seasonal effect on the rate of exceedances, and on a concentration of the highest excesses during the second half of the year, but it does not greatly affect estimates of overall return levels;
- (c) there is some evidence of *trend* but based almost entirely on having several of the highest exceedances in the last 5 years of the data, so it is difficult to draw conclusions.

Overall, it appears that the influence of a few large observations is affecting the whole analysis, making it difficult both to obtain return level estimates and to assess such things as trend and seasonality. However, the coverage intervals quoted in Table 7 show a reasonable consistency over different methods of analysis, and the fact that they are so wide appears to be a fair reflection of the information contained in the data about these extreme quantiles.

Finally, we make some comparisons with the approach based on annual maxima. Fitting the generalized extreme value distribution to the 35 annual maxima led to $\hat{k} = -0.32$ with a quantile-quantile plot very similar to Fig. 7, the two largest observations being well below their predicted values under the model. The profile likelihoods for return levels are considerably flatter than those of Fig. 9, presumably because of the smaller number of data points being used. Smith (1990) has details of this. Hosking *et al.* (1985) proposed estimation by probability weighted moments, but it is far from clear that this solves any of the basic problems with the data. Although the threshold method is hard to implement satisfactorily, the annual maximum method gives far worse results with these data.

10. DISCUSSION

The generalized Pareto distribution may be taken as the basis of a very broad modelling approach to high level exceedances. It can be used in conjunction with regression techniques (Sections 3–6) and extensions to deal with serially dependent and seasonal data (Sections 7–9) cover most of the practical situations met with univariate series.

Our emphasis has been with modelling rather than the more specific and thorny problem of predicting extreme values and estimating extreme quantiles. As shown by the detailed example in Section 9, this can be very difficult, though it should be said that this data set appears to be unusually troublesome. The analysis raises questions about appropriate methods of inference, especially the comparison among different methods of constructing confidence intervals and the comparison with Bayesian techniques. Overall, however, the confidence intervals quoted fairly represent the information in the data. We have also dealt rather lightly with the selection of the threshold, which is a difficult theoretical problem with some analogies with selecting a smoothing parameter in nonparametric regression or density estimation. Other possible modelling problems include multivariate series and the use of more specific models, such as Markov chains, for serial dependence. The whole area is rich with interesting problems, both theoretical and practical. In this paper we have studied some of these problems and proposed our own solutions, but even for the examples that we have studied there are many open problems and possibilities for alternative analyses.

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APPENDIX A: SOME LIKELIHOOD CALCULATIONS

Suppose that the Y_i , $i = 1, \dots, N$ are independently distributed with generalized Pareto distributions with parameters k_i and σ_i , and that in each case $k_i < \frac{1}{2}$. Suppose also that $\sigma_i = \sigma(\mathbf{x}_i^T \boldsymbol{\beta})$ and $k_i = k(\mathbf{z}_i^T \boldsymbol{\gamma})$, where $\mathbf{x}_i$ and $\mathbf{z}_i$ are vectors of known covariates of respective lengths $p+1$ and $q+1$. Suppose in addition that the Y_i may be upper truncated at random, and say that $\delta_i = 0$ if y_i is the observed value of the random variable Y_i , but that $\delta_i = 1$ if y_i is a lower bound for the true but unobserved Y_i .

Throughout this appendix, $0 \leq s, t \leq q$, $0 \leq u, v \leq p$, and σ', k' and so forth are the derivatives of σ and k with respect to their arguments, while $w = y/\sigma$.

The contribution to the overall log-likelihood $L = \sum l_i$ made by Y_i is

$$l_i(y_i; \sigma_i, k_i) = \begin{cases} -y_i/\sigma_i - \log \sigma_i + \delta_i \log \sigma_i & k_i = 0 \\ \left(\frac{1}{k_i} - 1 \right) \log \left(1 - k_i \frac{y_i}{\sigma_i} \right) - \log \sigma_i + \delta_i \left(\log \sigma_i + \log \left(1 - k_i \frac{y_i}{\sigma_i} \right) \right) & k_i \neq 0, \end{cases}$$

$$(\sigma_i > 0, y_i > 0, 1 - k_i y_i / \sigma_i > 0).$$

The score vector has components

$$\frac{\partial L}{\partial \gamma_s} = \sum_{i=1}^N z_{is} k_i' \frac{\partial l_i(y_i; \sigma_i, k_i)}{\partial k_i},$$

$$\frac{\partial L}{\partial \beta_u} = \sum_{i=1}^N x_{iu} \sigma_i' \frac{\partial l_i(y_i; \sigma_i, k_i)}{\partial \sigma_i},$$

where

$$\frac{\partial l_i(y; \sigma, k)}{\partial k} = \begin{cases} -\frac{1}{k^2} \log(1 - kw) + \left(1 - \delta_i - \frac{1}{k} \right) \frac{w}{1 - kw} & k \neq 0 \\ (1 - \delta_i) w - \frac{1}{2} w^2 & k = 0 \end{cases}$$

$$\frac{\partial l_i(y; \sigma, k)}{\partial \sigma} = \frac{1}{\sigma} \frac{w + \delta_i - 1}{1 - kw}.$$

The components of the observed information matrix I_θ are

$$-\frac{\partial^2 L}{\partial \gamma_s \partial \gamma_t} = - \sum_{i=1}^N z_{is} z_{it} \left\{ k_i'^2 \frac{\partial^2 l_i(y_i; \sigma_i, k_i)}{\partial k_i^2} + k_i'' \frac{\partial l_i(y_i; \sigma_i, k_i)}{\partial k_i} \right\},$$

$$-\frac{\partial^2 L}{\partial \gamma_s \partial \beta_u} = - \sum_{i=1}^N z_{is} x_{iu} k_i' \sigma_i' \frac{\partial^2 l_i(y_i; \sigma_i, k_i)}{\partial k_i \partial \sigma_i},$$

$$-\frac{\partial^2 L}{\partial \beta_u \partial \beta_v} = - \sum_{i=1}^N x_{iu} x_{iv} \left\{ \sigma_i'^2 \frac{\partial^2 l_i(y_i; \sigma_i, k_i)}{\partial \sigma_i^2} + \sigma_i'' \frac{\partial l_i(y_i; \sigma_i, k_i)}{\partial \sigma_i} \right\},$$

where

$$\frac{\partial^2 l_i(y; \sigma, k)}{\partial k^2} = \begin{cases} \frac{2}{k^3} \log(1 - kw) + \frac{2w}{k^2(1 - kw)} + \left(1 - \frac{1}{k} - \delta_i \right) \left(\frac{w}{1 - kw} \right)^2 & k \neq 0 \\ w^2(1 - \delta_i - \frac{2}{3}w) & k = 0, \end{cases}$$

$$\frac{\partial^2 l_i(y; \sigma, k)}{\partial k \partial \sigma} = \frac{w}{\sigma} \frac{w + \delta_i - 1}{(1 - kw)^2},$$

$$\frac{\partial^2 l_i(y; \sigma, k)}{\partial \sigma^2} = \frac{1}{\sigma^2} \frac{1 - 2w + kw^2 - \delta_i}{(1 - kw)^2}.$$

The components of the expected information matrix i_θ are

$$E\left(\frac{-\partial^2 L}{\partial \gamma_s \partial \gamma_t}\right) = \sum_{i=1}^N z_{is} z_{it} k_i'^2 i_{kk}(y_i; \sigma_i, k_i),$$

$$E\left(\frac{-\partial^2 L}{\partial \gamma_s \partial \beta_u}\right) = \sum_{i=1}^N z_{is} x_{iu} k_i' \sigma_i' i_{k\sigma}(y_i; \sigma_i, k_i),$$

$$E\left(\frac{-\partial^2 L}{\partial \beta_u \partial \beta_v}\right) = \sum_{i=1}^N x_{iu} x_{iv} \sigma_i'^2 i_{\sigma\sigma}(y_i; \sigma_i, k_i),$$

where the i_{kk} and so forth are given by the following expressions:

$$i_{kk}(y; \sigma, k) = \frac{1}{(1-k)(1-2k)} \{2 - \delta_i(2 + 2w + w^2 - 4kw - 3kw^2 + 2k^2w^2)(1-kw)^{1/k-2}\};$$

$$i_{k\sigma}(y; \sigma, k) = -\frac{1}{\sigma(1-k)(1-2k)} \{1 - \delta_i(1 + w + 2kw)(1-kw)^{1/k-2}\};$$

$$i_{\sigma\sigma}(y; \sigma, k) = \frac{1}{\sigma^2(1-2k)} \{1 - \delta_i(1 - kw)^{1/k-2}\}.$$

APPENDIX B: FITTING THE GENERALIZED PARETO DISTRIBUTION IN GLIM

Suppose that no data are censored and there is a single value of k . Let μ denote the vector of fitted values $(\sigma_1, \dots, \sigma_n)^T$, and η the vector of linear predictors $(\mathbf{x}_1^T \beta, \dots, \mathbf{x}_n^T \beta)^T$. In GLIM notation, calculate fitted values $\%FV = \mu = \sigma(\eta)$, deviance increments $\%DI = t/n + 2\{\log \mu + (1 - 1/k) \log(1 - ky/\mu)\}$ and link function derivative $\%DR = d\sigma^{-1}(\mu)/d\mu$. The constant t is chosen to ensure that every component of the deviance is positive. Use expected information to calculate the variance function, by $\%VA = V = (1 - 2k)\mu^2$, and declare y variate $\%YV = \mu + V(y - \mu)/\mu(\mu - ky)$. If k is positive, the distribution has finite support and the restriction $1/k > y/\mu$ holds. If violated, this restriction can be enforced by repeatedly multiplying μ by a scaling factor $r > 1$ until $\max\{y_i/\mu_i\} < 1/k$, and then using the resulting μ as starting values for the iterative weighted least squares algorithm. Good results are obtained with $r = 1.2$. If k is initially taken to be negative and increased through the range of interest, the values of $\hat{\beta}_1, \dots, \hat{\beta}_p$ typically change little from one step to the next, and $\hat{\beta}_0$ increases with k . The result is a fast and reliable algorithm for estimation of β , minor alterations to which are needed to estimate parameters from censored data. GLIM programs for the purpose can be obtained from the first author.

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DISCUSSION OF THE PAPER BY DAVISON AND SMITH

C. W. Anderson (University of Sheffield): This is an important paper. It deals with problems which are important outside statistics, and it offers fresh thinking on them. It can be seen as part of a move over the past 6 or 7 years to break out from the Gumbel framework which has dominated statisticians' thinking about extremes since the 1940s. The paper reinterprets, in the light of recent probabilistic advances, *ad hoc* models already used in hydrology and builds linear modelling methodology into them to deepen them in a profound way. The resulting methods are more faithful to the underlying structure of the data, and are powerful and flexible in applications.

Of the many issues discussed in the paper, I would like to focus on the authors' general ideas about extremes in time series.

The paper's main thesis is that it can be fruitful to model large values in terms of the times of exceedance of a high level and the amounts by which the level is overtopped. It is fair to observe, however, that this idea is not completely realized in the paper, since only methods for the fitting of *peak* excesses are developed, leaving data on other excesses within a cluster unused. As information about extremes is almost always scarce, it would be highly desirable to press these subsidiary data into use too (especially as they embody other aspects of extreme behaviour—such as the durations of extreme excursions and the aggregate excess over a threshold—which can be important in their own right). This is bound to be difficult, since we can expect excesses within a cluster to be dependent.

Some information relevant to the modelling of what occurs within clusters is given by complete convergence limit theorems, which describe, by means of a two-dimensional point process representation, the limiting properties of the times of occurrence of large values and also their magnitudes, after suitable normalization. The most general such theorem current is due to Hsing (1987) and covers stationary sequences satisfying a mild long-range dependence condition. Though Hsing's formulation is not directly in terms of exceedances of a threshold, it is possible to deduce various properties of exceedances and excesses from his results.

One rather striking deduction from them, for example, is that the distribution of excess for a randomly chosen exceedance *has exactly the same limiting distribution as that of a cluster peak*. This apparent paradox is a consequence of length-biased sampling: a randomly chosen exceedance has a disproportionate chance of coming from a large cluster, and in large clusters there tend to be large excesses. The special structure of the limiting point process ensures that the two effects work together exactly to give the equality.

A statistical consequence is that data on excesses from *all* exceedances should be usable for estimation of the generalized Pareto distribution (GPD) of peak excess, provided that the method of estimation does not rely on knowledge of dependence within a cluster. This is relevant to the estimation of the marginal distribution of exposures towards the end of Section 6, in which clustering and dependence were ignored (and the estimation was ultimately based on likelihood). It would be interesting to hear the authors' views on whether, in general, likelihood estimation of the GPD distribution of peak excess from data on all excesses, neglecting dependence, will give satisfactory results.

The theoretical equality of the two distributions for excess also suggests that a comparison of their two empirical estimates could be used to check applicability of the limiting distributional assumptions. If the estimates differ markedly then doubt is cast on the assumptions.

This, and the fact that, according to the limit theorem, excesses within a cluster must conform to a particular kind of pattern, raises the general question of whether models suggested by asymptotic arguments will always have a sufficiently rich structure to represent non-asymptotic reality. Here, as